

FREE VIBRATION OF ANISOTROPIC COMPOSITE THIN-WALLED BEAMS OF CLOSED CROSS-SECTION CONTOUR

O. SONG AND L. LIBRESCU

*Department of Engineering Science and Mechanics, Virginia Polytechnic Institute and
State University, Blacksburg, Virginia 24061-0219, U.S.A.*

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This paper focuses on the formulation of the dynamic problem of laminated composite thick- and thin-walled, single-cell beams of arbitrary cross-section and on the investigation of their associated free vibration behavior. The structural model considered herein incorporates a number of non-classical effects. These are related to the exotic properties characterizing the advanced composite material structures; namely, anisotropy, heterogeneity and transverse shear flexibility, and with the incorporation of the non-uniform torsional model.

The results obtained in this paper, related to the modelling of thick and thin-walled beams (TWB) made of advanced composite materials and the clarification of the role played by a number of non-classical effects, are expected to be useful toward obtaining better predictions of the vibrational behavior of TWB structures, in general, and of dynamic response and flutter instability of aeronautical/aerospace structures, in particular.

1. INTRODUCTION

Because of their obvious advantages, such as high strength to weight ratio and high stiffness to weight ratio, fiber reinforced laminated composite thick- and thin-walled beam structures are likely to play a great role in the design of the present and, especially, of future generations of aeronautical and aerospace, as well as of naval constructions.

In addition to the above-mentioned advantages, the various elastic and structural couplings, which are the result of the directional nature of composite materials and of the ply-stacking sequence, could successfully be exploited to enhance the response characteristics of aerospace and naval vehicles.

In order to be able to determine the behavior of these structures, consistent theories and pertinent analytical tools, allowing one to determine their response to complex static and dynamic excitations, are required. Along these lines, one of the goals of this paper consists of the accurate modelling of the dynamics of laminated anisotropic composite thick- and thin-walled beams of closed cross-section contour.

Another goal consists of the study of the vibrational behavior of cantilevered anisotropic thick- and thin-walled beams and of the assessment of the influence played by the various effects which are incorporated. Among them, the material anisotropy induced by the directional nature of fibrous composites is shown to play a great role toward the enhancement of eigenvibration characteristics of thin- and thick-walled beams.

Despite the evident practical importance of the problem, research work in this field appears to be extremely scarce. To the best knowledge of the authors, a number of results concerning the structural dynamic behavior of thin-walled beams of closed cross-section

contour modelled within various levels of refinement and devoted to metallic type structures are contained in papers [1-3], and devoted to composite ones in references [4-11]. A review of the literature in this field can be found in reference [11].

Although in the title of the paper the term "thin" was used in association with this kind of structure, it has to be made clear that the theory developed here can also accommodate the case of *thick-walled* beams (for which case $h_{max}/b \geq 0.1$, where h_{max} denotes the maximum thickness of the beam and b is a typical cross-sectional dimension). However, throughout the paper, the standard terminology of "thin-walled beams" will be used to designate also the beams which from the thinness requirement point of view, are not thin (in the sense that their dimensions do not fulfill the condition $h_{max}/b < 0.1$). Moreover, as it was previously shown in the case of the theory of anisotropic shells and plates [12] (the conclusions will remain valid in this case, too), the necessity of incorporating the transverse shear effect is not conditioned by the thinness characteristic only, but also by the degree of flexibility in transverse shear exhibited by the constituent materials of the beam structure.

2. ASSUMPTIONS AND KINEMATICS

The case of single-cell, thin-walled beams of arbitrary closed-section is considered. The points of the beam structures are referred to a 3-D co-ordinate system (x, y, z) , where the z (spanwise) co-ordinate coincides with a straight unspecified reference axis, while x and y are the cross-section co-ordinates, the origin of which lies on the reference axis (see Figure 1). The equations of thin-walled beams are derived by adopting a number of statements, as follows.

- (i) The original cross-sections of the beam remain unchanged. This statement implies that the cross-sections do not deform in their own planes [1-10, 13-20].
- (ii) In contrast to the uniform torsion model (referred to as Saint-Venant torsion), in which framework the rate of twist ϕ' is considered to be constant, in the case of

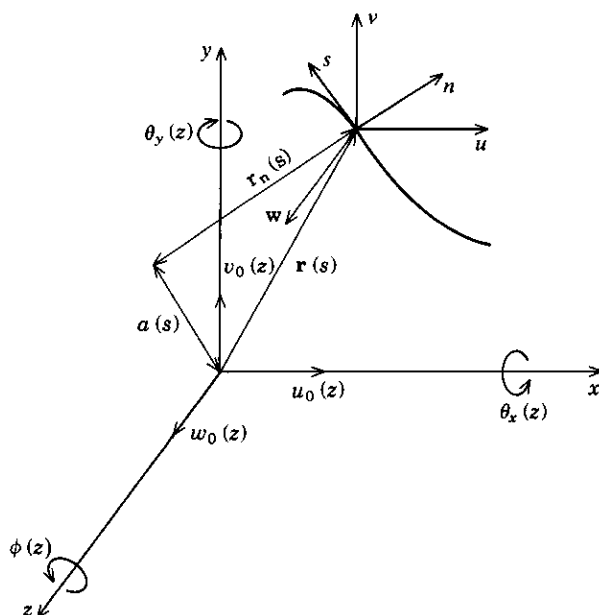


Figure 1. Co-ordinate systems and kinematic variables.

non-uniform torsion (referred to as restrained-torsion), the rate of twist is considered a function of the spanwise co-ordinate z , i.e., $\phi' \equiv \phi'(z)$ [14]. Here and in the following equations, differentiation with respect to the axial co-ordinate is denoted by primes ($(\cdot)' \equiv d(\cdot)/dz$). $\phi'(z)$ constitutes a measure of the warping restraint effect. With regard to the distribution of the warping displacement along the mid-line of the wall (referred to as primary warping), a distribution $F_\omega(s)$ similar to the one associated with its uniform-torsion model counterpart is postulated [14–18].

- (iii) In addition to the primary warping effect, the warping of the cross-section points off the mid-line contour (referred to as the secondary warping) is incorporated [19].
- (iv) The transverse shear effect is incorporated [16]. Moreover, it is stipulated that transverse shear strains γ_{xz} and γ_{yz} are uniform for each cross-section (in the sense that $\gamma_{xz} \equiv \gamma_{xz}(z)$ and $\gamma_{yz} \equiv \gamma_{yz}(z)$).
- (v) Besides the previous assumptions of a kinematic nature, the following static one is postulated. In its terms, the circumferential stress resultant N_{ss} (i.e., the hoop stress resultant) is assumed to be negligibly small when compared to the remaining ones [16].

By virtue of the statement (ii) above, the primary warping function $F_\omega(s)$ assumes the form [14]:

$$F_\omega(s) = \int_0^s [r_n(\bar{s}) - \psi] d\bar{s}, \quad (1)$$

where the torsional function ψ is

$$\psi = \frac{\oint r_n(\bar{s}) d\bar{s}}{\oint d\bar{s}} \left(\equiv \frac{2A_c}{\beta} \right), \quad (2)$$

while $\int_0^s r_n(\bar{s}) d\bar{s} = \Omega(s)$ denotes the sectorial area [13].

In these equations, A_c denotes the cross-sectional area of the beam bounded by the mid-line contour; s denotes the arc-length measured along the circumferential co-ordinate, the origin of which is arbitrarily but conveniently chosen, $\bar{s}$ is a dummy co-ordinate associated with the s co-ordinate; $\oint(\cdot) d\bar{s}$ denotes the integral along the closed mid-line contour; $r_n(s)$ is a geometric quantity (see Figure 1) given by

$$r_n(s) = x(s) \frac{dy}{ds} - y(s) \frac{dx}{ds}, \quad (3)$$

while $\beta = \oint d\bar{s}$ denotes the circumferential contour length.

By virtue of assumptions (i)–(iv) and in order to reduce the 3-D elasticity problem of the associated thin-walled beam theory to an equivalent 1-D one, the components of the 3-D displacement vector are expressed as [10, 17]:

$$\begin{aligned} u(x, y, z; t) &= u_0(z; t) - y\phi(z; t), \\ v(x, y, z; t) &= v_0(z; t) + x\phi(z; t), \\ w(x, y, z; t) &= w_0(z; t) + \theta_x(z; t) \left[y(s) - n \frac{dx}{ds} \right] \\ &\quad + \theta_y(z; t) \left[x(s) + n \frac{dy}{ds} \right] - \phi'(z; t) [F_\omega(s) + na(s)]. \end{aligned} \quad (4)$$

It should be remarked that x and y are the co-ordinates of a point on the mid-line contour and are, therefore, functions of the contour co-ordinate s , whereas $F_w(s)$ and $na(s)$ play the roles of the primary and secondary warping functions, respectively. In equations (4)

$$\begin{aligned}\theta_x(z; t) &= \gamma_{yz}(z; t) - v'_0(z; t), & \theta_y(z; t) &= \gamma_{xz}(z; t) - u'_0(z; t), \\ a(s) &= -y(s) \frac{dy}{ds} - x(s) \frac{dx}{ds}.\end{aligned}\quad (5)$$

In the light of equations (4) and (5), it is readily seen that the statement of the cross-section non-deformability (implying $\varepsilon_{xx}=0$; $\varepsilon_{yy}=0$ and $\gamma_{xy}=0$ and, consequently, also $\varepsilon_{nn}=\varepsilon_{ss}=\gamma_{sn}=0$), and the continuity requirement of w along the mid-line contour (i.e., $\oint (\partial w / \partial s) ds = 0$) are fulfilled.

In equation (4), n denotes the co-ordinate in the thickness direction, while t denotes the time variable. In addition, $u_0(z; t)$, $v_0(z; t)$ and $w_0(z; t)$ denote the rigid body translations along the x , y and z axes; while $\theta_x(z; t)$, $\theta_y(z; t)$ and $\phi(z; t)$, the rigid body rotations about the x - and y -axes and the twist about the z -axis, respectively, constitute the basic unknown functions of the problem.

3. CONSTITUTIVE EQUATIONS

Consider the case of composite TWB's constituted by the superposition of a finite number N of individually homogeneous layers. Consider that the material of each constituent layer is linearly elastic and anisotropic, and that the layers are perfectly bonded.

The 3-D constitutive equations of a generally orthotropic elastic material could be expressed in matrix form as

$$\begin{Bmatrix} \sigma_{ss} \\ \sigma_{zz} \\ \sigma_{nn} \\ \sigma_{zn} \\ \sigma_{ns} \\ \sigma_{sz} \end{Bmatrix} = \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{13} & 0 & 0 & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{23} & 0 & 0 & \bar{Q}_{26} \\ \bar{Q}_{13} & \bar{Q}_{23} & \bar{Q}_{33} & 0 & 0 & \bar{Q}_{36} \\ 0 & 0 & 0 & \bar{Q}_{44} & \bar{Q}_{45} & 0 \\ 0 & 0 & 0 & \bar{Q}_{45} & \bar{Q}_{55} & 0 \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{36} & 0 & 0 & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_{ss} \\ \varepsilon_{zz} \\ \gamma_{nn} \\ \gamma_{zn} \\ \gamma_{ns} \\ \gamma_{sz} \end{Bmatrix}.\quad (6)$$

In equation (6), $\bar{Q}_{ij}$ denote the transformed elastic coefficients associated with the k th layer in the global co-ordinate system of the structure, while $\gamma_{pr}=2\varepsilon_{pr}$ when $p \neq r$, and ε_{ij} denote the components of the strain tensor. The constitutive equations (6) exhibit a 3-D dependence. They are reduced to an equivalent 1-D dependence in two steps. The first step, yielding the 2-D constitutive equations (CE) (referred to as local CE's), consists of the integration of their original 3-D form throughout the laminate thickness, while the second one, resulting in the 1-D form, consists of the integration of the previous form of CE's along the mid-line contour of the cross-section of the beam.

In light of statements (i) and (v), the local CE's expressed in terms of the strain measures, are as follows.

The stress resultants are

$$\begin{Bmatrix} N_{zz} \\ N_{sz} \end{Bmatrix} = \begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{21} & K_{22} & K_{23} & K_{24} \end{bmatrix} \begin{Bmatrix} \varepsilon_{zz}^0 \\ \gamma_{sz}^0 \\ \phi' \\ \varepsilon_{zz}^n \end{Bmatrix},\quad (7)$$

The transverse shear stress resultant is

$$N_{zn} = A_{44} \gamma_{zn}, \quad (8)$$

and the stress couples are

$$\begin{Bmatrix} L_{zz} \\ L_{sz} \end{Bmatrix} = \begin{bmatrix} K_{41} & K_{42} & K_{43} & K_{44} \\ K_{51} & K_{52} & K_{53} & K_{54} \end{bmatrix} \begin{Bmatrix} \varepsilon_{zz}^0 \\ \gamma_{sz}^0 \\ \phi' \\ \varepsilon_{zz}^n \end{Bmatrix}. \quad (9)$$

In equations (7) and (9), the K_{ij} denote the modified local stiffness coefficients, whose expressions are displayed in Appendix A. The strain measures involved in these equations are

$$\begin{aligned} \varepsilon_{zz}^0 &= w'_0 + x(s)\theta'_y + y(s)\theta'_x - F_w(s)\phi'', & \varepsilon_{zz}^n &= \theta'_y \frac{dy}{ds} - \theta'_x \frac{dx}{ds} - a(s)\phi'', \\ \gamma_{sz}^0 &= [u'_0 + \theta_y] \frac{dx}{ds} + (v'_0 + \theta_x) \frac{dy}{ds} + 2 \frac{A_c}{\beta} \phi', & \gamma_{zn} &= [u'_0 + \theta_y] \frac{dy}{ds} - [v'_0 + \theta_x] \frac{dx}{ds}, \end{aligned} \quad (10)$$

where ε_{zz}^n and ε_{zz}^0 denote the axial strain components associated with the secondary and primary warping, respectively, while γ_{sz}^0 and γ_{zn} denote the membrane and transverse shear strain, respectively.

In equations (7)–(9) both the stress resultants, the stress couples and the strain measures exhibit a time dependence and a 2-D spatial one (the dependence being on the s and z co-ordinates).

4. THE EQUATIONS OF MOTION AND BOUNDARY CONDITIONS

In the previous sections, the kinematic equations and local constitutive equations of thin-walled composite beams of closed cross-section have been displayed. In order to obtain the equations of motion and the associated boundary conditions, Hamilton's variational principle of the 3-D elastic theory is used. By virtue of this principle, of all displacements $v_i \equiv v_i(x, y, z; t)$ that satisfy the boundary conditions (BC's) $v_i = \bar{v}_i$ over Ω_b and that fulfill also the condition $\delta v_i = 0$ at $t = t_0$ and $t = t_1$, the actual ones fulfill the variational equation (see reference [12])

$$\delta J = \int_{t_0}^{t_1} dt \left[\int_{\tau} \sigma_{ij} \delta \varepsilon_{ij} d\tau - \delta K - \int_{\Omega_e} \bar{x}_i \delta v_i d\Omega - \int_{\tau} \rho H_i \delta v_i d\tau \right] = 0, \quad (11)$$

where

$$K = \frac{1}{2} \int_{\tau} \rho (\dot{v}_i \dot{v}_i) d\tau \quad (12)$$

denotes the kinetic energy.

In addition, t_0 and t_1 denote two arbitrary instants of time; $d\tau (\equiv dn ds dz)$ denotes the differential volume element; $s_i (\equiv \sigma_{ij} n_j)$ denote the components of the stress vector on a surface element of the undeformed body displaying the outward normal components n_i ; the H_i denote the components of the body forces; Ω_e denotes the external area of the body over which the stresses are prescribed; ρ denotes the mass density; and an underlined sign identifies a prescribed quantity while δ denotes the variation sign. In equations (11)

and (12), the Einsteinian summation convention applies to repeated indices, where Latin indices range from 1 to 3. In the same equations, $\{v_1, v_2, v_3\} \equiv \{u, v, w\}$, while $(x_1, x_2, x_3) \equiv (x, y, z)$.

Consideration of equation (4) in conjunction with constitutive equation, assuming the variations δu_0 , δv_0 , δw_0 , $\delta \theta_x$, $\delta \theta_y$ and $\delta \phi$ as independent and arbitrary, performing the integrations over the s - and n -directions, and introduction of the definitions for the 1-D forces and moments,

$$\begin{aligned} T_z(z; t) &\equiv \oint N_{zz} ds, & Q_x(z; t) &= \oint \left(N_{sz} \frac{dx}{ds} + N_{zn} \frac{dy}{ds} \right) ds, \\ Q_y(z; t) &= \oint \left(N_{sz} \frac{dy}{ds} - N_{zn} \frac{dx}{ds} \right) ds, & M_x(z; t) &= \oint \left(y N_{zz} - L_{zz} \frac{dx}{ds} \right) ds, \\ M_y(z; t) &= \oint \left(x N_{zz} + L_{zz} \frac{dy}{ds} \right) ds, & M_z(z; t) &= 2 \oint N_{sz} \psi ds, \end{aligned} \quad (13)$$

and

$$B_\omega(z; t) = \oint [F_\omega(s) N_{zz} + a(s) L_{zz}] ds,$$

yields, from equations (11) and (12), the 1-D equations of motion and the associated BC's. In equation (13), T_z , Q_x and Q_y denote the axial and shear forces (in the x and y directions), while M_x , M_y , M_z and B_ω denote the moments (about the x -, y - and z -axes) as well as the bimoment global quantities. Discarding the terms associated with the external loadings and body forces, the obtained equations of motions are

$$\begin{aligned} \delta u_0: \quad & \underline{\underline{M_y''}} + \underline{\underline{Q_x'}} = \underline{\underline{I_1}} - \underline{\underline{I_6'}}, \\ \delta v_0: \quad & \underline{\underline{M_x''}} + \underline{\underline{Q_y'}} = \underline{\underline{I_2}} - \underline{\underline{I_8'}}, \\ \delta w_0: \quad & T_z' = I_3, \\ \delta \phi: \quad & \underline{\underline{B_\omega''}} + \underline{\underline{M_z'}} = \underline{\underline{I_4}} - \underline{\underline{I_9'}}, \\ \delta \theta_y: \quad & \underline{\underline{M_y' - Q_x}} = \underline{\underline{I_5}}, \\ \delta \theta_x: \quad & \underline{\underline{M_x' - Q_y}} = \underline{\underline{I_7}}. \end{aligned} \quad (14)$$

and the associated static BC's at the ends ($z = 0, L$) of the beam became

$$\begin{aligned} \delta u_0: \quad & \underline{\underline{M_y' + Q_x}} + \underline{\underline{I_6}} = \underline{\underline{Q_x}}, \\ \delta v_0: \quad & \underline{\underline{M_x' + Q_y}} = \underline{\underline{I_8}} = \underline{\underline{Q_y}}, \\ \delta w_0: \quad & T_z = \underline{\underline{I_3}}, \\ \delta \theta_y \text{ or } \delta u_0': \quad & M_y = \underline{\underline{M_y}}, \\ \delta \theta_x \text{ or } \delta v_0': \quad & M_x = \underline{\underline{M_x}}, \\ \delta \phi: \quad & B_\omega' + M_z + I_9 = \underline{\underline{M_x}}, \\ \delta \phi': \quad & \underline{\underline{B_\omega}} = \underline{\underline{B_\omega}}, \end{aligned} \quad (15a)$$

and the geometric BC's become

$$\begin{aligned} u_0 &= \underline{u}_0, & v_0 &= \underline{v}_0, & w_0 &= \underline{w}_0, \\ \underline{\theta}_y &= \underline{\underline{\theta}}_y \quad \text{or} \quad \underline{u}'_0 = \underline{u}'_0, & \underline{\theta}_x &= \underline{\underline{\theta}}_x \quad \text{or} \quad \underline{v}'_0 = \underline{v}'_0, \\ \phi &= \underline{\phi}, & \underline{\phi}' &= \underline{\underline{\phi}}'. \end{aligned} \quad (15b)$$

In these equations, I_i denote the inertia terms while $\underline{Q}_x$, $\underline{Q}_y$, $\underline{T}_z$, $\underline{M}_y$, $\underline{M}_x$, $\underline{M}_z$ and $\underline{B}_\omega$ denote the prescribed static quantities, their expressions being displayed in Appendices B and C, respectively.

In equations (14) and (15) the single and double solid underlines identify the terms pertaining to the infinitely rigid and flexible in transverse shear theories, respectively. When the former theory is considered, implying

$$\theta_y(z) \rightarrow -u'_0(z), \quad \theta_x(z) \rightarrow -v'_0(z), \quad (16)$$

the terms identified by the double solid underlines have to be discarded. Conversely, when the transverse shear deformable theory is adopted, the terms affected by a single solid line have to be dropped. In addition, the two options arising in several of the BC's correspond to the shear deformable and infinitely-rigid-in-transverse-shear beam theories, respectively.

For the shear deformable thin-walled beam theory, a similar form of the equations of motion was obtained in references [7, 16, 17]. Elimination of $\underline{Q}_x$ and $\underline{Q}_y$ in equations (14) of the shear deformable theory results in a system of equations of motion which is pertinent to its infinitely-rigid-in-transverse-shear counterpart (obtainable also from equation (14)). Towards obtaining this form of the equations of motion, the relationships

$$I_6 \rightleftharpoons -I_5, \quad I_8 \rightleftharpoons -I_7, \quad (17)$$

are useful. They arise when in the relationships (17) the replacements $u'_0 \rightleftharpoons -\theta_y$ and $v'_0 \rightleftharpoons -\theta_x$ are performed. Moreover, the free warping counterpart of equations (14) and (15) can be obtained by discarding therein the quantities affected by the dotted line.

It should be remarked that, consistent with the seven boundary conditions to be prescribed at each end of the beam, a 14th order governing system is obtained. However, its free warping counterpart entails prescription of only six BC's and, as a result, a 12th order governing system is obtained. Another remark concerns the fact that the difference in order of the systems of governing equations, associated with the shear deformable and infinitely rigid in transverse shear thin-walled beam theories, is immaterial. A similar feature is encountered in the theory of solid anisotropic beams, where the incorporation of transverse shear flexibility is not accompanied by an increase of the order of the associated governing equations (see [21, 22]).

5. THE GOVERNING SYSTEM

The most convenient formulation of the dynamic governing system and of the associated BC's is in terms of the kinematic quantities u_0 , v_0 , w_0 , θ_x , θ_y and ϕ . As in the theory of plates and shells (see, e.g., reference [12]), this representation could be obtained by expressing the stress resultants and stress couples entering the equations of motion in terms of the unknown kinematic variables. After lengthy calculations, the dynamic governing system of differential equations associated with the transverse shear deformation theory

of anisotropic and heterogeneous TWB's is expressed as:

$$\begin{aligned}
 \delta u_0: \quad & a_{41}w_0'' + a_{42}\theta_y'' + a_{43}\theta_x'' + a_{44}(\theta_y' + u_0'') + a_{45}(\theta_x' + v_0'') + a_{46}\phi''' + a_{47}\phi'' = b_1\ddot{u}_0 - b_2\ddot{\phi}, \\
 \delta v_0: \quad & a_{51}w_0'' + a_{52}\theta_y'' + a_{53}\theta_x'' + a_{54}(\theta_y' + u_0'') + a_{55}(\theta_x' + v_0'') + a_{56}\phi''' + a_{57}\phi'' = b_1\ddot{v}_0 + b_3\ddot{\phi}, \\
 \delta w_0: \quad & a_{11}w_0'' + a_{12}\theta_y'' + a_{13}\theta_x'' + a_{14}(\theta_y' + u_0'') + a_{15}(\theta_x' + v_0'') + a_{16}\phi''' + a_{17}\phi'' \\
 & = b_1\ddot{w}_0 + b_3\ddot{\theta}_y + b_2\ddot{\theta}_x - b_7\phi', \\
 \delta\theta_y: \quad & a_{21}w_0'' + a_{22}\theta_y'' + a_{23}\theta_x'' + a_{24}(\theta_y' + u_0'') + a_{25}(\theta_x' + v_0'') + a_{26}\phi''' + a_{27}\phi'' \\
 & - a_{41}w_0' - a_{42}\theta_y' - a_{43}\theta_x' - a_{44}(\theta_y + u_0') - a_{45}(\theta_x + v_0') - a_{46}\phi'' - a_{47}\phi' \\
 & = b_3\ddot{w}_0 + (b_5 + b_{15})\ddot{\theta}_y + (b_6 + b_{13})\ddot{\theta}_x - (b_9 + b_{17})\phi', \\
 \delta\theta_x: \quad & a_{31}w_0'' + a_{32}\theta_y'' + a_{33}\theta_x'' + a_{34}(\theta_y' + u_0'') + a_{35}(\theta_x' + v_0'') + a_{36}\phi''' + a_{37}\phi'' \\
 & - a_{51}w_0' - a_{52}\theta_y' - a_{53}\theta_x' - a_{54}(\theta_y + u_0') - a_{55}(\theta_x + v_0') - a_{56}\phi'' - a_{57}\phi' \\
 & = b_2\ddot{w}_0 + (b_6 + b_{13})\ddot{\theta}_y + (b_4 + b_{14})\ddot{\theta}_x - (b_8 + b_{16})\phi', \\
 \delta\phi: \quad & -a_{61}w_0''' - a_{62}\theta_y''' - a_{63}\theta_x''' - a_{64}(\theta_y'' + u_0''') - a_{65}(\theta_x'' + v_0''') - a_{66}\phi'''' - a_{67}\phi''' - a_{71}w_0'' \\
 & + a_{72}\theta_y'' + a_{73}\theta_x'' + a_{74}(\theta_y' + u_0'') + a_{75}(\theta_x' + v_0'') + a_{76}\phi''' + a_{77}\phi'' \\
 & = b_3\ddot{v}_0 - b_2\ddot{u}_0 + (b_4 + b_5)\ddot{\phi} + b_7\ddot{w}_0 + (b_9 + b_{17})\ddot{\theta}_y + (b_8 - b_{16})\ddot{\theta}_x - (b_{10} + b_{18})\phi''.
 \end{aligned} \tag{18}$$

6. THE BOUNDARY CONDITIONS

For cantilevered TWB's, the BC's at the root are entirely geometric, whereas at the tip they are entirely static. Expressed in terms of the kinematic variables, the homogeneous BC's are as follows. At the root of the beam ($z = 0$):

$$u_0 = v_0 = w_0 = \theta_y = \theta_x = \phi = \phi' = 0, \tag{19}$$

and at the tip of the beam ($z = L$):

$$\begin{aligned}
 \delta u_0: \quad & a_{41}w_0' + a_{42}\theta_y' + a_{43}\theta_x' + a_{44}(\theta_y + u_0') + a_{45}(\theta_x + v_0') + a_{46}\phi'' + a_{47}\phi' = 0, \\
 \delta v_0: \quad & a_{51}w_0' + a_{52}\theta_y' + a_{53}\theta_x' + a_{54}(\theta_y + u_0') + a_{55}(\theta_x + v_0') + a_{56}\phi'' + a_{57}\phi' = 0, \\
 \delta w_0: \quad & a_{11}w_0' + a_{12}\theta_y' + a_{13}\theta_x' + a_{14}(\theta_y + u_0') + a_{15}(\theta_x + v_0') + a_{16}\phi'' + a_{17}\phi' = 0, \\
 \delta\theta_y: \quad & a_{21}w_0' + a_{22}\theta_y' + a_{23}\theta_x' + a_{24}(\theta_y + u_0') + a_{25}(\theta_x + v_0') + a_{26}\phi'' + a_{27}\phi' = 0, \\
 \delta\theta_x: \quad & a_{31}w_0' + a_{32}\theta_y' + a_{33}\theta_x' + a_{34}(\theta_y + u_0') + a_{35}(\theta_x + v_0') + a_{36}\phi'' + a_{37}\phi' = 0, \\
 \delta\phi: \quad & -a_{61}w_0'' - a_{62}\theta_y'' - a_{63}\theta_x'' - a_{64}(\theta_y' + u_0'') - a_{65}(\theta_x' + v_0'') - a_{66}\phi''' - a_{67}\phi'' + a_{71}w_0' + a_{72}\theta_y' \\
 & + a_{73}\theta_x' + a_{74}(\theta_y + u_0') + a_{75}(\theta_x + v_0') + a_{76}\phi'' + a_{77}\phi' \\
 & = b_7\ddot{w}_0 + (b_9 + b_{17})\ddot{\theta}_y + (b_8 - b_{16})\ddot{\theta}_x - (b_{10} + b_{18})\phi', \\
 \delta\phi': \quad & a_{61}w_0' + a_{62}\theta_y' + a_{63}\theta_x' + a_{64}(\theta_y + u_0') + a_{65}(\theta_x + v_0') + a_{66}\phi'' + a_{67}\phi' = 0.
 \end{aligned} \tag{20}$$

In equations (18)–(20), a_{ij} ($i, j = \overline{1, 7}$) and b_i ($i = \overline{1, 18}$) denote the stiffness and mass coefficient, respectively. For the sake of completeness, they are displayed in Appendices D and E, respectively.

It could be concluded that in this case of general anisotropy and heterogeneity the governing system and the associated BC's exhibit a complete coupling between the various modes; i.e., warping (primary and secondary), bending (vertical and lateral), twist and transverse shearing. The assessment of the influence of these couplings on the various problems and their proper exploitation should constitute an important task toward a rational design of these structures, and of the proper use of the exotic material characteristics (i.e., anisotropy and ply-stacking sequence) generating these couplings.

7. SPECIAL CASES

Two important special cases of the previously obtained governing equation system will be considered. These ones arise in the cases of the ply-angle distributions labelled as configurations A and B, respectively. They are:

$$A: \theta(y) = \theta(-y), \quad B: \theta(y) = -\theta(-y).$$

It should be remarked in this context that the ply angles (measured from the positive s -axis (see Figure 2)) could be obtained through the usual filament winding technology.

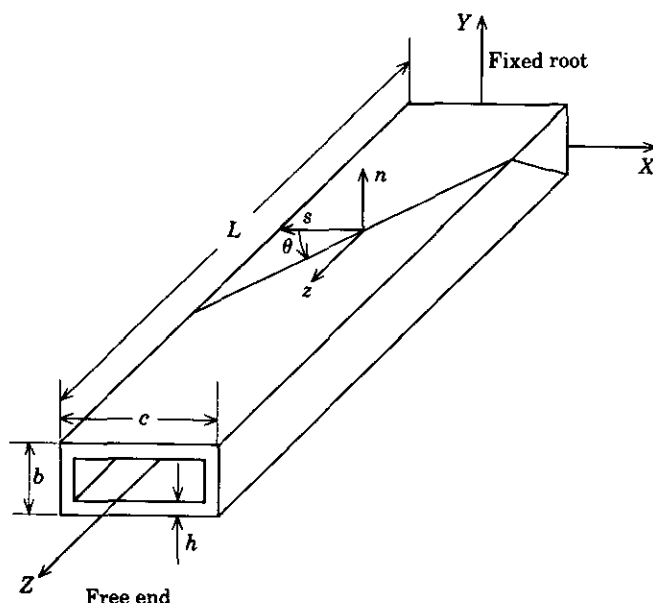


Figure 2. The geometry of the box beam (not to scale): $h = 0.04$ in, $b = 0.2$ in, $c = 1.0$ in, $L = 10.0$ in.

Configuration A results in structural decoupling between bending and torsion, manifested both in the governing equations (including the inertia terms) and the boundary conditions. The decoupling yields two subsystems of coupled governing equations and boundary conditions. A former system consisting of two coupled equations (with the associated BC's) governs the extension-twist vibrations, while the latter one, consisting of four coupled equations (with the associated BC's) governs the bending-transverse shear vibrations. These equations, also incorporating the warping restraint effect (primary and secondary), are as follows.

(A)₁ The equations governing the extension-twist motion (of the sixth order) are

$$\begin{aligned}\delta w_0: \quad & a_{11}w_0'' + a_{17}\phi'' = b_1\ddot{w}_0, \\ \delta\phi: \quad & -a_{66}\phi''' + a_{77}\phi'' + a_{71}w_0'' = (b_4 + b_5)\ddot{\phi} - (b_{10} + b_{18})\ddot{\phi}',\end{aligned}\quad (21)$$

with the associated BC's at $z=L$,

$$\begin{aligned}\delta w_0: \quad & a_{11}w_0' + a_{17}\phi' = 0, \\ \delta\phi: \quad & -a_{66}\phi''' + a_{71}w_0' + a_{77}\phi' = -(b_{10} + b_{18})\ddot{\phi}', \\ \delta\phi': \quad & -a_{66}\phi'' = 0,\end{aligned}\quad (22)$$

and with the associated BC's at $z=0$,

$$w_0 = \phi = \phi' = 0.$$

(A)₂ The equations governing the bending-transverse shear vibrations (of the eighth order) are

$$\begin{aligned}\delta u_0: \quad & a_{43}\theta_x'' + a_{44}(u_0'' + \theta_y') = b_1\ddot{u}_0, \\ \delta v_0: \quad & a_{52}\theta_y'' + a_{55}(v_0'' + \theta_x') = b_1\ddot{v}_0, \\ \delta\theta_y: \quad & a_{22}\theta_y'' + a_{25}(v_0'' + \theta_x') - a_{44}(u_0' + \theta_y') - a_{43}\theta_x' = (b_5 + b_{15})\ddot{\theta}_y, \\ \delta\theta_x: \quad & a_{33}\theta_x'' + a_{34}(u_0'' + \theta_y') - a_{55}(v_0' + \theta_x) - a_{52}\theta_y' = (b_4 + b_{14})\ddot{\theta}_x,\end{aligned}\quad (23)$$

with the associated BC's at $z=L$,

$$\begin{aligned}\delta u_0: \quad & a_{43}\theta_x' + a_{44}(u_0' + \theta_y) = 0, \\ \delta v_0: \quad & a_{52}\theta_y' + a_{55}(v_0' + \theta_x) = 0, \\ \delta\theta_y: \quad & a_{22}\theta_y' + a_{25}(v_0' + \theta_x) = 0, \\ \delta\theta_x: \quad & a_{33}\theta_x' + a_{34}(u_0' + \theta_y) = 0,\end{aligned}\quad (24)$$

and the associated BC's at $z=0$,

$$u_0 = v_0 = \theta_x = \theta_y = 0.$$

Within configuration B, the governing system and the associated BC's split exactly into two groups, (B)₁ and (B)₂, each one exhibiting the following coupling mechanisms.

(B)₁ The system of equations (of the sixth order) governing extension – lateral bending – lateral transverse shear motions:

$$\begin{aligned}\delta w_0: \quad & a_{11}w_0'' + a_{14}(u_0'' + \theta_y') = b_1\ddot{w}_0, \\ \delta u_0: \quad & a_{41}w_0'' + a_{44}(u_0'' + \theta_y') = b_1\ddot{u}_0, \\ \delta\theta_y: \quad & -a_{41}w_0' - a_{44}(u_0' + \theta_y) + a_{22}\theta_y'' = (b_5 + b_{15})\ddot{\theta}_y,\end{aligned}\quad (25)$$

with the associated BC's at $z=L$,

$$\begin{aligned}\delta w_0: \quad & a_{11}w_0' + a_{14}(u_0' + \theta_y) = 0, \\ \delta u_0: \quad & a_{41}w_0' + a_{44}(u_0' + \theta_y) = 0, \\ \delta\theta_y: \quad & a_{22}\theta_y' = 0,\end{aligned}\quad (26)$$

and at $z=0$,

$$w_0 = u_0 = \theta_y = 0.$$

(B)₂ The system of equations (of eighth order) governing the twist – vertical bending – vertical transverse shear motions:

$$\begin{aligned}\delta\phi: \quad & -\underline{a_{66}\phi''''} + a_{73}\theta_x'' + a_{77}\phi'' = (b_4 + b_5)\ddot{\phi} - (b_{10} + b_{18})\ddot{\phi}'', \\ \delta\theta_x: \quad & a_{33}\theta_x'' + a_{37}\phi'' - a_{55}(v_0' + \theta_x) - \underline{a_{56}\phi''} = (b_4 + b_{14})\ddot{\theta}_x, \\ \delta v_0: \quad & a_{55}(v_0'' + \theta_x') + \underline{a_{56}\phi'''} = b_1\ddot{v}_0,\end{aligned}\quad (27)$$

with the associated BC's at $z = L$,

$$\begin{aligned}\delta\phi: \quad & -\underline{a_{66}\phi'''} + a_{73}\theta_x' + a_{77}\phi' = -(b_{10} + b_{18})\ddot{\phi}', \\ \delta v_0: \quad & a_{55}(v_0' + \theta_x) + \underline{a_{56}\phi''} = 0, \\ \delta\theta_x: \quad & a_{33}\theta_x' + a_{37}\phi' = 0, \\ \delta\phi': \quad & \underline{a_{66}\phi''} = 0,\end{aligned}\quad (28)$$

and at $z=0$,

$$\phi = v_0 = \theta_x = \phi' = 0.$$

The Saint-Venant torsional model counterpart of sets (A)₁ and (B)₂, (i.e., the ones discarding the warping restraint effect) was obtained previously in reference [18] without the display of the associated boundary conditions. In papers [7, 18], the configurations A and B are referred to as *circumferentially uniform stiffness* (CUS) and *circumferentially asymmetric stiffness* (CAS) configurations, respectively. A further decoupling of equation (23) and of the associated BC's (24) occurs in the case of cross-ply TWB structures. In this case, the coupling rigidities $a_{43} (\equiv a_{34}) = a_{52} (\equiv a_{25}) = 0$ and, as a result, the vertical bending – transverse shear vibrations decouple from the lateral bending – lateral transverse shear vibrations. In this case, each subsystem is characterized by a fourth order governing system and correspondingly, by two BC's to be fulfilled at $z=0, L$.

The specialization of equations (25) and (27) and of the BC's (equations (26) and (28)) for the same case of cross-ply structures which yields $a_{14} (\equiv a_{41}) = a_{73} (\equiv a_{37}) = a_{56} = 0$, results in the same decoupling of governing equations as in the configuration A case. A similar decoupling also occurs in the case of TWB's symmetrically laminated of laminae of transversely isotropic materials (the plane of isotropy being parallel to the s - z surface).

It is seen that, in these special cases, the difference between configurations A and B becomes immaterial. In any of the previous cases (general or special), in order to approach the undamped free vibration problem, the unknown functions denoted generically as $\mathcal{F}(z, t)$ are represented in the form

$$\mathcal{F}(z, t) = \mathcal{F}(z) \exp(i\omega t), \quad i = \sqrt{-1}. \quad (29)$$

In this case the problem is reduced to an eigenvalue problem which is exactly solved by using the Laplace transform technique developed in references [22–24].

8. NUMERICAL ILLUSTRATIONS

The case of a box beam of rectangular cross-section is considered. The considered geometric parameters of the box beam are $h = 0.04$ in, $b = 0.2$ in, $c = 1.0$ in and $L = 10.0$ in (see Figure 2), whereas the physical characteristics of the material (graphite/epoxy) in the on-axis configuration are $E_1 = 30 \times 10^6$ psi, $E_2 = E_3 = 0.75 \times 10^6$ psi, $G_{13} = G_{23} = 0.37 \times 10^6$ psi, $G_{12} = 0.45 \times 10^6$ psi, $\mu_{12} = \mu_{23} = \mu_{13} = 0.25$, and $\rho = 14.3 \times 10^{-5}$ lb s²/in⁴.

The numerical illustrations concern only the structural configuration A which, from the point of view of the induced elastic couplings is more advantageous in the design of helicopter blades and tilt rotor aircraft (see e.g. references [7, 8]).

In Figure 3, the effect of tailoring on the eigenfrequencies of the lateral and vertical (plunging) vibrational modes (referred to, in helicopter terminology, as lead-lag bending and flap bending, respectively) is displayed. These results incorporate the (secondary) warping and the transverse shear deformation effects.

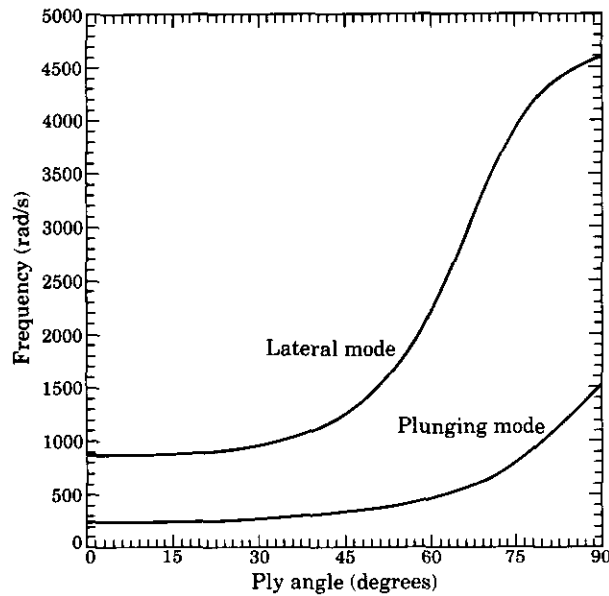


Figure 3. Variation of the fundamental eigenfrequencies *vs.* ply angle for lateral and plunging vibration modes.

In Figures 4 and 5, the effects of transverse shear flexibility on the lateral and vertical vibration eigenfrequencies, for various ply angles, are displayed.

In Figure 6, the variation of the fundamental eigenfrequencies for the coupled extension-twist vibrations with the variation of the ply angle is depicted.

In Table 1, the influence of the secondary warping effect and of the ply-angle variation on the fundamental and second eigenfrequencies in the vertical bending vibration mode is displayed.

Finally, in Tables 2 and 3 is shown the influence of the transverse shear deformability on the first two eigenfrequencies associated with the lateral and vertical vibration modes, respectively.

9. DISCUSSION AND CONCLUSIONS

A study of the free vibration of cantilevered composite aircraft wing structures was accomplished. For this purpose, an advanced model for the wing structure, namely that of an anisotropic composite thin-walled beam of closed contour, was considered. The results emerging from Figure 3 reveal that, for larger ply angles, higher bending eigenfrequencies are obtained. This fact is justifiable by comparing the magnitudes of the bending stiffness quantities for various ply angles. Figure 3 also reveals that the eigenfrequencies of the lateral bending mode are higher than those of the plunging vibrational mode. This result can be predicted by comparing the bending stiffnesses for each bending vibration

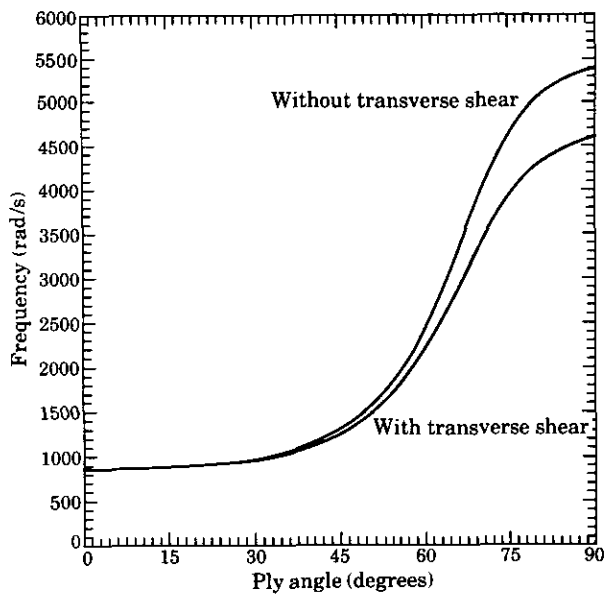


Figure 4. The effect of the transverse shear on the fundamental eigenfrequencies of the lateral bending mode.

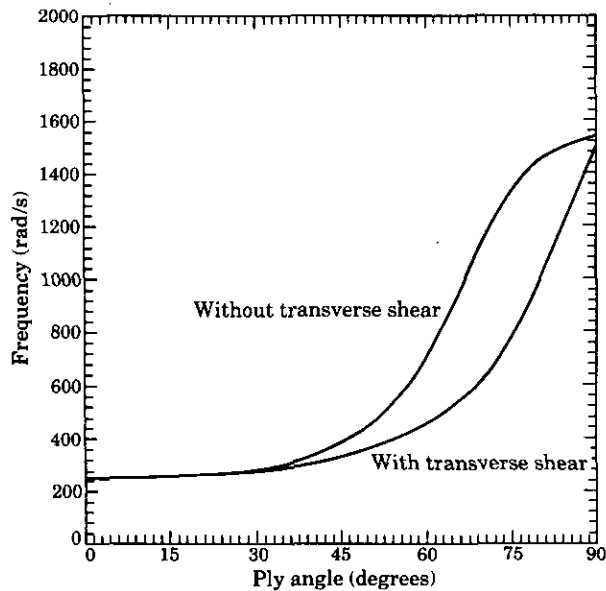


Figure 5. The effect of the transverse shear on the fundamental eigenfrequencies of the plunging bending mode.

mode. Due to the geometry of the beam cross-section considered here, the magnitude of the area moment of inertia about the y -axis is larger than that about the x -axis.

In Figures 4 and 5, the effect of the transverse shear deformability on the eigenfrequencies of the two bending vibrational modes was illustrated. It is revealed that the incorporation of the transverse shear flexibility results in lower eigenfrequencies than in their rigid-in transverse-shear counterparts. A similar trend, obtained also in references [4, 5, 9], verifies the one also present in the refined theories of plates and shells (see, e.g., reference

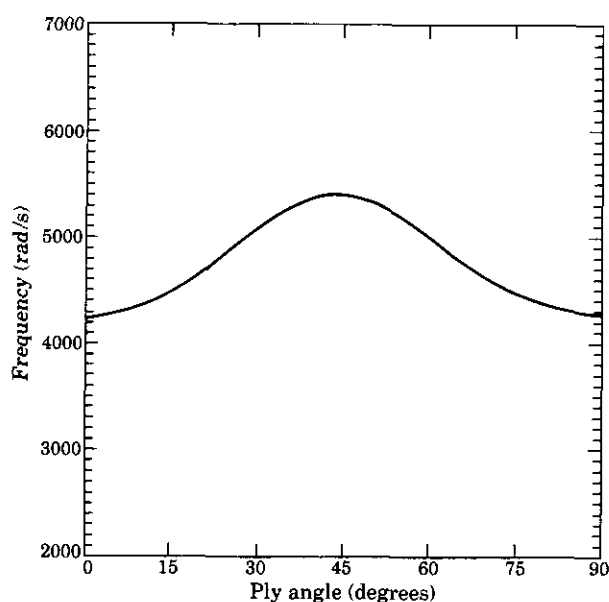


Figure 6. Variation of the fundamental eigenfrequencies vs. ply angle for the extension-torsion coupled mode.

TABLE 1

Variation of the eigenfrequencies vs. ply angle by including (II) and discarding (I) the secondary warping effect (rad/s)

Ply angle (degrees)	Fundamental		Second	
	I	II	I	II
0	248	249	1548	1558
15	251	253	1564	1577
30	264	269	1646	1676
45	306	318	1906	1985
60	425	443	2672	2786
75	749	765	4853	4934
90	1493	1502	8473	8502

TABLE 2

Effect of the transverse shear on the first two eigenfrequencies (rad/s) for the lateral bending mode: I, without transverse shear; II, with transverse shear (values in parentheses are percentage differences)

Ply angle (degrees)	Fundamental		Second	
	I	II	I	II
15	886	880 (-0.6)	5 463	4 383 (-19.8)
30	955	949 (-0.6)	5 898	4 657 (-21.0)
45	1 276	1 236 (-9.0)	7 877	5 490 (-40.5)
60	2 399	2 182 (-9.0)	14 804	7 554 (-49.0)
75	4 598	3 916 (-14.8)	28 380	12 333 (-56.5)

TABLE 3

Effect of the transverse shear on the first two eigenfrequencies (rad/s) for the plunging bending mode: I, without transverse shear; II, with transverse shear (values in parentheses are percentage differences)

Ply angle (degrees)	Fundamental		Second	
	I	II	I	II
0	250	249 (-0.4)	1564	1558 (-0.4)
15	254	253 (-0.4)	1585	1577 (-0.5)
30	277	269 (-2.9)	1728	1676 (-3.0)
45	373	318 (-14.7)	2335	1985 (-15.0)
60	693	443 (-36.1)	4337	2786 (-35.8)
75	1315	765 (-41.8)	8230	4934 (-40.0)
90	1530	1502 (-1.8)	9576	8502 (-11.2)

[12]). In Tables 2 and 3, the strong effect of the transverse shear on the lateral and vertical bending eigenfrequencies is illustrated. Table 2 reveals that, in the case of the lateral bending vibration mode, the effect of the transverse shear is more prominent for the second frequencies than for the first ones. For the plunging vibration mode, this trend was not detected, in the sense that in both the fundamental and second eigenfrequencies, the transverse shear flexibility exhibits almost the same influence (see Table 3).

TABLE 4

Comparison of the eigenfrequencies (rad/s) of the torsional mode obtained within FW and WR beam models, $\theta=90^\circ$ (values in parentheses are percentage differences)

	Fundamental	Second
FW	4 239	12 717
WR	4 640 (8.6)	14 595 (13.0)

With regard to the warping inhibition, its effect, as revealed in Table 4 for the two considered cases, results in an increase of 13% of the eigenfrequencies of the torsional vibration mode. This implies that the warping restraint effect actually provides additional torsional stiffness. Table 4 also reveals that the effect of warping restraint is more severe in the higher modes than in the lower ones.

In Table 5, the variation of the eigenfrequencies in the axial vibration mode for two different ply angles is illustrated. Large variations in the eigenfrequencies for the two

TABLE 5

Eigenfrequencies (rad/s) in the axial vibration mode for two ply angles (values in parentheses are percentage differences)

Ply angle (degrees)	Fundamental	Second
0	11 749	35 247
90	72 007 (513.0)	216 021 (513.0)

different ply angle cases are detected. The powerful effect of the structural tailoring is clearly revealed in these numerical results.

The effect of the secondary warping effect on the eigenfrequencies of the plunging bending mode is emphasized in Table 1. The numerical results reveal that the inclusion of the secondary warping effect yields an increase of bending eigenfrequencies. This fact is explainable, having in view that the secondary warping provides both an extra bending stiffness as well as an extra torsional one. Its contribution is also dependent on the thickness ratio of the considered structure. In this numerical illustration, the inclusion of the secondary warping effect results in an increase of only 5% of the eigenfrequency. It is believed, however, that for other cases, e.g., for relatively thick-walled beams; for special cross-section shapes (e.g., Neuber-tube configuration, for which case the primary warping becomes immaterial), as well as for the closed section beam counterpart of the case reported in reference [25], *larger contributions of this effect could be detected.*

Last but not least, it should be acknowledged that, due to the complexity of the equations involved, the symbolic manipulation programs MACSYMA were extensively used in their derivation [26].

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APPENDIX A

$$\begin{aligned}
 K_{11} &= A_{22} - A_{12}^2/A_{11}, & K_{43} &= 2K_{24}A_c/\beta, \\
 K_{12} &= A_{26} - A_{12}A_{16}/A_{11} = K_{21}, & K_{44} &= D_{22} - B_{12}^2/A_{11}, \\
 K_{13} &= 2K_{12}A_c/\beta, & K_{51} &= B_{26} - B_{16}A_{12}/A_{11}, \\
 K_{14} &= B_{22} - A_{12}B_{12}/A_{11} = K_{41}, & K_{52} &= B_{66} - B_{16}A_{16}/A_{11}, \\
 K_{22} &= A_{66} - A_{16}^2/A_{11}, & K_{53} &= 2K_{52}A_c/\beta, \\
 K_{23} &= 2K_{22}A_c/\beta, & K_{54} &= D_{26} - B_{12}B_{16}/A_{11}, \\
 K_{24} &= B_{26} - A_{16}B_{12}/A_{11} = K_{42},
 \end{aligned}$$

where A_{ij} , B_{ij} , and D_{ij} denote local stretching, stretching-bending coupling and bending rigidity quantities, respectively.

APPENDIX B

$$\begin{aligned}
 I_1 &= \oint (\ddot{u}_0 - y\ddot{\phi})m_0 ds, & I_2 &= \oint (\ddot{v}_0 + x\ddot{\phi})m_0 ds, \\
 I_3 &= \oint (\ddot{w}_0 + x\ddot{\theta}_y + y\ddot{\theta}_x - F_\omega\ddot{\phi}')m_0 ds, & I_4 &= \oint \{(x^2 + y^2)\ddot{\phi} - y\ddot{u}_0 + x\ddot{v}_0\}m_0 ds,
 \end{aligned}$$

$$\begin{aligned}
I_5 &= \oint [x\ddot{w}_0 + x^2\ddot{\theta}_y + xy\ddot{\theta}_x - xF_\omega(s)\ddot{\phi}']m_0 \, ds + \oint \left(\frac{dy}{ds} \frac{dy}{ds} \ddot{\theta}_y - \frac{dx}{ds} \frac{dy}{ds} \ddot{\theta}_x - \frac{dy}{ds} a(s)\ddot{\phi}' \right) m_2 \, ds, \\
I_6 &= \oint [-x\ddot{w}_0 + x^2\ddot{u}'_0 + xy\ddot{v}'_0 + xF_\omega(s)\ddot{\phi}']m_0 \, ds + \oint \left(\frac{dy}{ds} \frac{dy}{ds} \ddot{u}'_0 - \frac{dx}{ds} \frac{dy}{ds} \ddot{v}'_0 + \frac{dy}{ds} a(s)\ddot{\phi}' \right) m_2 \, ds, \\
I_7 &= \oint [y\ddot{w}_0 + y^2\ddot{\theta}_x + xy\ddot{\theta}_y - yF_\omega(s)\ddot{\phi}']m_0 \, ds + \oint \left[\frac{dx}{ds} \frac{dx}{ds} \ddot{\theta}_x - \frac{dx}{ds} \frac{dy}{ds} \ddot{\theta}_y + \frac{dx}{ds} a(s)\ddot{\phi}' \right] m_2 \, ds, \\
I_8 &= \oint [-y\ddot{w}_0 + y^2\ddot{v}'_0 + xy\ddot{u}'_0 + yF_\omega(s)\ddot{\phi}']m_0 \, ds + \oint \left[\frac{dx}{ds} \frac{dx}{ds} \ddot{v}'_0 - \frac{dx}{ds} \frac{dy}{ds} \ddot{u}'_0 - \frac{dx}{ds} a(s)\ddot{\phi}' \right] m_2 \, ds, \\
I_9 &= \oint [-F_\omega(s)\ddot{w}_0 - xF_\omega\ddot{\theta}_y - yF_\omega(s)\ddot{\theta}_x + F_\omega(s)^2\ddot{\phi}']m_0 \, ds \\
&\quad + \oint \left[\frac{dx}{ds} a(s)\ddot{\theta}_x - \frac{dy}{ds} a(s)\ddot{\theta}_y + a^2(s)\ddot{\phi}' \right] m_2 \, ds,
\end{aligned}$$

where

$$(m_0, m_2) = \sum_{k=1}^N \int_{h_{(k-1)}}^{h_{(k)}} \rho_{(k)}(1, n^2) \, dn.$$

APPENDIX C

$$\begin{aligned}
Q_x &\equiv \oint \sum_{k=1}^N \int_{h_{(k)}} \mathcal{Q}_x \, dn \, ds, & M_y &\equiv \oint \sum_{k=1}^N \int_{h_{(k)}} [x\mathcal{Q}_z + \frac{dy}{ds} n\mathcal{Q}_z] \, dn \, ds, \\
Q_y &\equiv \oint \sum_{k=1}^N \int_{h_{(k)}} \mathcal{Q}_y \, dn \, ds, & M_x &\equiv \oint \sum_{k=1}^N \int_{h_{(k)}} [y\mathcal{Q}_z + \frac{dx}{ds} n\mathcal{Q}_z] \, dn \, ds, \\
I_z &\equiv \oint \sum_{k=1}^N \int_{h_{(k)}} \mathcal{Q}_z \, dn \, ds, & M_z &\equiv \oint \sum_{k=1}^N \int_{h_{(k)}} [-y\mathcal{Q}_x + x\mathcal{Q}_y] \, dn \, ds, \\
B_\omega &\equiv - \oint \sum_{k=1}^N \int_{h_{(k)}} [F_\omega \mathcal{Q}_z + na\mathcal{Q}_z] \, dn \, ds,
\end{aligned}$$

where

$$\int_{h_{(k)}} (\cdot) \, dn \equiv \int_{h_{(k-1)}}^{h_{(k)}} (\cdot) \, dn.$$

APPENDIX D

$$\begin{aligned}
a_{11} &= \oint K_{11} \, ds, & a_{12} &= \oint \left(xK_{11} + K_{14} \frac{dy}{ds} \right) ds, & a_{13} &= \oint \left(yK_{11} - K_{14} \frac{dx}{ds} \right) ds, \\
a_{14} &= 3K_{12} \frac{dx}{ds} ds, & a_{15} &= \oint K_{12} \frac{dy}{ds} ds, & a_{16} &= - \oint [K_{11}F_\omega(s) + K_{14}a] ds,
\end{aligned}$$

$$\begin{aligned}
a_{17} &= \oint K_{13} \, ds, & a_{22} &= \oint \left[K_{11}x^2 + 2xK_{14} \frac{dy}{ds} + K_{44} \frac{dy}{ds} \frac{dy}{ds} \right] ds, \\
a_{23} &= \oint \left[K_{11}xy - xK_{14} \frac{dx}{ds} + yK_{14} \frac{dy}{ds} - K_{44} \frac{dx}{ds} \frac{dy}{ds} \right] ds, \\
a_{24} &= \oint \left(xK_{12} \frac{dx}{ds} + K_{24} \frac{dx}{ds} \frac{dy}{ds} \right) ds, & a_{25} &= \oint \left(xK_{12} \frac{dy}{ds} + K_{24} \frac{dy}{ds} \frac{dy}{ds} \right) ds, \\
a_{26} &= - \oint \left[xK_{11}F_\omega + xK_{14}a + F_\omega K_{14} \frac{dy}{ds} + K_{44}a \frac{dy}{ds} \right] ds, & a_{27} &= \oint \left[xK_{13} + K_{43} \frac{dy}{ds} \right] ds, \\
a_{33} &= \oint \left(K_{11}y^2 - 2yK_{14} \frac{dx}{ds} + K_{44} \frac{dx}{ds} \frac{dx}{ds} \right) ds, & a_{34} &= \oint \left(yK_{12} \frac{dx}{ds} - K_{24} \frac{dx}{ds} \frac{dx}{ds} \right) ds, \\
a_{35} &= \oint \left(yK_{12} \frac{dy}{ds} - K_{24} \frac{dx}{ds} \frac{dy}{ds} \right) ds, \\
a_{36} &= - \oint \left(yK_{11}F_\omega + yK_{14}a - F_\omega K_{14} \frac{dx}{ds} - K_{44}a \frac{dx}{ds} \right) ds, \\
a_{37} &= \oint \left[yK_{13} - K_{43} \frac{dx}{ds} \right] ds, & a_{44} &= \oint \left(K_{22} \frac{dx}{ds} \frac{dx}{ds} + A_{44} \frac{dy}{ds} \frac{dy}{ds} \right) ds, \\
a_{45} &= \oint \left(K_{22} \frac{dx}{ds} \frac{dy}{ds} - A_{44} \frac{dx}{ds} \frac{dy}{ds} \right) ds, & a_{46} &= - \oint \left(F_\omega K_{21} \frac{dx}{ds} + K_{24}a \frac{dx}{ds} \right) ds, \\
a_{47} &= \oint \left[K_{23} \frac{dx}{ds} \right] ds, & a_{55} &= \oint \left(K_{22} \frac{dy}{ds} \frac{dy}{ds} + A_{44} \frac{dx}{ds} \frac{dx}{ds} \right) ds, \\
a_{56} &= - \oint \left(F_\omega K_{21} \frac{dy}{ds} + K_{24}a \frac{dy}{ds} \right) ds, \\
a_{57} &= \oint \left[K_{23} \frac{dy}{ds} \right] ds, & a_{66} &= \oint \left[K_{11}F_\omega^2 + 2K_{14}F_\omega a + K_{44}a^2 \right] ds, \\
a_{67} &= - \oint \left[K_{13}F_\omega + K_{43}a \right] ds, & a_{77} &= \oint \left[2 \frac{A_c}{\beta} K_{23} \right] ds.
\end{aligned}$$

The remaining rigidity quantities not appearing here are obtained from the present ones by using the symmetry relations $a_{ij} = a_{ji}$ ($i, j = 1, 7$).

APPENDIX F

$$\begin{aligned}
&(b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9, b_{10}, b_{11}, b_{12}) \\
&= \oint m_0 \left(1, y, x, y^2, x^2, xy, F_\omega, yF_\omega, xF_\omega, F_\omega^2, \frac{dx}{ds}, \frac{dy}{ds} \right) ds. \\
&(b_{13}, b_{14}, b_{15}, b_{16}, b_{17}, b_{18}) = \oint m_2 \left(\frac{dx}{ds} \frac{dy}{ds}, \left(\frac{dx}{ds} \right)^2, \left(\frac{dy}{ds} \right)^2, a \frac{dx}{ds}, a \frac{dy}{ds}, a^2 \right) ds.
\end{aligned}$$